Lab 1 – FAICT Description

Colby Scroggins

Old Dominion University

CS411W

Professor Hosni

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1. Introduction

Internet users are subjected to rising levels of misinformation from online sources and platforms. Misinformation spreads rapidly across social media, news sites, and online communities, a problem that is heavily exacerbated by the rise of AI generated content. The scale of the issue is vast, affecting everyday web browsing and social media usage. Approximately 64% of U.S. adults report that misinformation causes major confusion. In addition to this, while roughly 94% of adults say they check the accuracy of their news sources, not all do so before sharing information with others (Gold, 2016; Cruz & Petrino, 2025). Consequently, 97% of those surveyed want stricter measures to limit the spread of misinformation (Cruz & Petrino, 2025).

The current process for combating this issue is often slow, confusing, and time-consuming. Manual validation requires extensive online searches and cross-referencing of sources, placing a heavy burden on the user. Existing content verification features offered by specific websites and platforms are limited in their scope and accessibility. These tools frequently suffer from platform exclusivity, popularity bias, and a primary focus on basic text verification. Current AI and chatbot offerings also fail to provide an intuitive, reliable, all-in-one solution. Manual verification is easily outpaced by the sheer volume of fake content that is being generated by AI tools.

Ultimately, users lack a universal and on-demand method for quickly verifying the information they encounter on the internet, and the burden of manual validating content results in many users skipping the process entirely. The required solution must be automated, fast, and capable of providing clear, evidence-based credibility assessments. Therefore, FAICT - the Fact

AI Checking Tool, is being proposed to help fill in these gaps and make browsing the web a more reliable experience across all web-based content.

2. Product Description

FAICT is an automated content verification system designed for real-time claim validation across text, audio, and video media formats. The FAICT infrastructure operates by analyzing the provided input, extracting claims, and then cross-referencing the claims against trusted sources and reliable information bases. For each claim that FAICT analyses, the system outputs a verdict based on the credibility of the claim alongside the sources used to reach the verdict. Simultaneously, the system alerts the user to whether the content is likely to have been generated by artificial intelligence. In reducing the friction between encountering and validating claims on the internet, FAICT's automation of the content verification process aims to mitigate the spread of misinformation.

2.1. Key Product Features and Capabilities

The frontend of FAICT's pipeline can accept inputs from a range of media including text, audio, and video. The system is able to accept non-text based media through the implementation of audio and video ingestion modules. When analyzing audio and video content, the system utilizes speech recognition modules to convert the claims into text transcripts. These transcripts are then processed in the same manner as a provided text-based input, in which claims are extracted, cross-referenced, and validated. The inclusion of speech-to-text modules and transcription generation allow for FAICT to provide real time verification of non-text based media such as podcasts, videos, and active conversations.

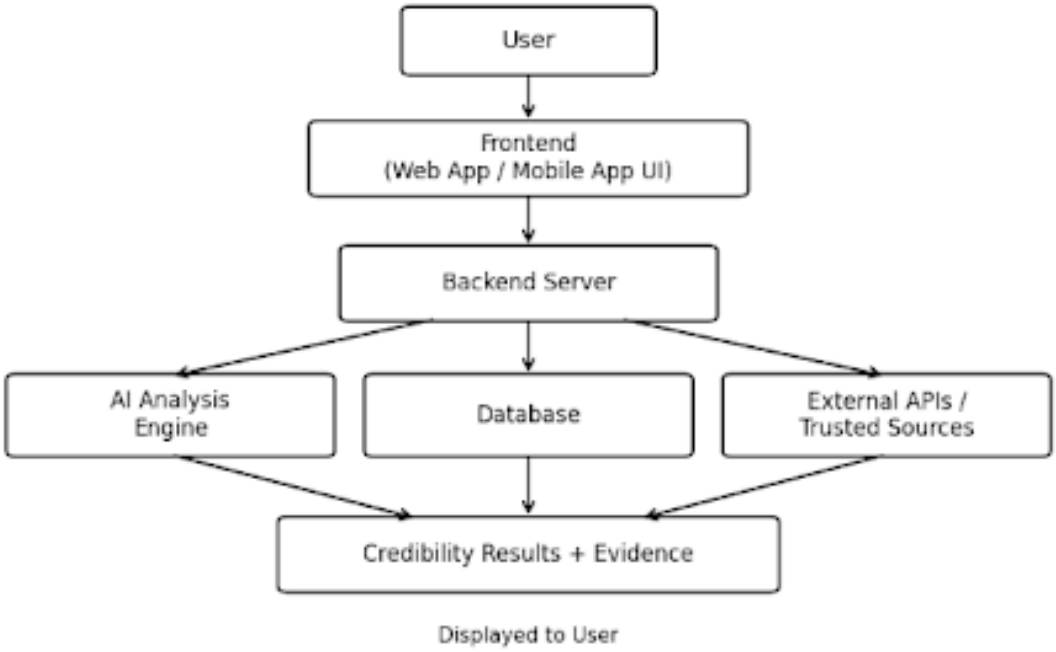
The validity and generative authenticity of text inputs, speech-to-text transcripts, and other media streams are evaluated concurrently once a claim is provided to the system. The first analysis pass determines the validity of the provided claim by conducting a web search to find corroborating or refuting evidence from a vast number of trusted sources and information bases. The data gathered during this pass is what allows for FAICT to provide a credibility assessment of the claims such as true, misleading, false, or uncertain, while supplementing this assessment with explanations and cited sources. The second pass employs an AI detection module to look for markers of AI generation in the provided input, such as statistical signifiers and AI watermarks.

The system is designed to be lightweight and easily accessible, providing the user with an intuitive application that automates the content verification process across a vast number of online sources. The product will be accessible through various means including the product's homepage, via a browser extension, and as a downloadable mobile application. The system's infrastructure ensures that backend processes like transcription, inference, and API requests scale to the user's needs to reduce latency and minimize disruptions during regular usage.

2.2. Major Components (Hardware/Software)

FAICT is a software-based content verification tool that is accessed via the internet. The software is being developed using a Python backend, HTML/CSS/JavaScript frontend interface, and a PostgreSQL or MongoDB database. The Major Functional Components Diagram outlines the structure of the system architecture:

Figure 1: Major Functional Components Diagram



The content verification process begins in the user realm, in which a piece of media - text, URLs, audio, video - is provided to the system via the frontend interface. The frontend serves as the user's entry point into the system's internal processing feature set and is presented to the user in the form of a responsive web application and downloadable mobile application. Once claims are extracted, analyzed, and corroborated or refuted, the frontend then presents the user with a complete assessment of the provided claims. The claim assessment page provides a credibility assessment, links to the sources used, and a determination of the generative authenticity of the claim.

The system's backend acts as a hub, connecting the frontend with the AI analysis engines, the system database architecture, external APIs, and other third party services employed by the system. Once the input is provided, the backend server delegates tasks to carry out the processing and analysis of the input data. The backend also includes the system database, where data such as user accounts, histories, claim extractions, and finalized validity assessments are stored.

User inputs are routed through the backend server to the AI analysis engine and external APIs for processing. This portion of the system covers tasks including claim extraction, speech-to-text, video transcription generation, and AI-generated content detection. Here, the data is processed and the system conducts the required web searches while simultaneously testing for signs of AI content generation. The results from the AI analysis engine and external APIs are extracted and combined to create the system's credibility assessments, which are then displayed to the user via the frontend. The table below details the intended feature set of FAICT:

Table 1: FAICT Feature Set

Feature	Description
User Input	Allows users to provide the system with input via text or url.
Claim Extraction	The system utilizes AI API modules to extract claims from the user's input source.
Evidence Retrieval	The system uses API calls to perform searches to cross reference the claims and retrieve corroborating or refuting evidence.
Results Page	The webpage displays the system's findings, offering a credibility rating alongside supporting evidence.
Account Page	Allows the user to create an account for subscription management and history

	retrieval.
Account History	Saves user history of previous claim analysis.

3. Identification of Case Study

FAICT is primarily being developed and curated towards college students, as this group often engages with a variety of online content ranging from social media to academic research. The case study for the FAICT prototype will involve a select group of undergraduate and graduate college students who will test the capability of FAICT in both of these use cases. Student researchers will test the system's capability to authenticate academic sources and materials, while other students will test the system's effectiveness during normal web browsing and social media usage. College students constitute an optimal user base for FAICT because their day-to-day activities involve working through dense academic materials where information accuracy is vital, as well as large amounts of unvetted content from a range of online sources in which misinformation and AI-generated content spread rapidly.

The primary use case for the FAICT prototype involves a student workflow centered around routine coursework and online media consumption. When the student encounters an unvetted article during research or a claim made in a video on social media, they can input the source directly into the FAICT frontend interface. The prototype will begin the information processing stage, utilizing modules to isolate and transcribe audio, extract claims from the text, and scan the media for signs of AI generation. Simultaneously, the prototype begins cross referencing the extracted claims to find corroborating or refuting evidence. The prototype then aggregates its findings and reaches a conclusion about the extracted claims, displaying them to the student alongside a credibility rating and cited primary sources. The flexibility of the browser

extension and mobile application allow for the student to validate claims in real time without a major disruption to their workflow.

Looking forward, FAICT seeks to expand its capabilities and user base beyond the goals outlined for the prototype and case study. The full capability release of FAICT will involve the integration of FAICT into enterprise software and university systems, allowing for student-submitted coursework to be fact-checked against trusted sources and known information bases, while also validating the generative authenticity of the assignments. Full integration into university systems will also allow for SSO into the FAICT ecosystem, securing student and faculty histories and notes across campus machines and mobile devices. By expanding the user base to include academic institutions and university libraries, FAICT also seeks to branch out into adjacent industries including digital journalism and broadcasting networks that rely on real-time content verification to operate safely and reliably.

4. Glossary

4.1. **AI:** Artificial Intelligence

4.2. **AI-Generated Content:** Text, images, audio, or video created or heavily modified by artificial intelligence tools.

4.3. **API:** Application Programming Interface

4.4. **Browser Extension:** A small software add-on that adds features or functionality to a web browser.

4.5. **Credibility Score:** A rating that estimates how trustworthy or reliable a claim appears based on supporting evidence.

4.6. **Disinformation:** False or misleading information that is intentionally created or shared to deceive people.

4.7. **Fact-Checking:** The process of verifying whether a claim is accurate by comparing it against reliable sources.

4.8. **FAICT:** the Fact AI Checking Tool

4.9. **GUI:** Graphical User Interface

4.10. **IDE:** Integrated Development Environment

4.11. **LLM:** Large Language Model

4.12. **Misinformation:** False or misleading information that may be shared regardless of whether there is intent to deceive.

4.13. **Speech-to-Text:** Technology that converts spoken audio into written text.

4.14. **SSO**: Single Sign-On

4.15. **Trusted Sources:** Reliable references, databases, publications, or organizations used to verify claims.

4.16. **Verification:** The process of confirming whether information is accurate, reliable, or supported by evidence.

4.17. **Video-to-Transcript:** The process of converting spoken content from a video into written text for analysis.

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